

Thermodynamic Arrow of Time: A Monotonically Increasing Entropy-Time Function

-From Multiplicative Entropy to Computational simulation

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ABSTRACT:

Each Moment Corresponds to an Entropy Value: A Rigorous Characterization of Time's Irreversibility In modern physics, the irreversibility of time is a central issue. Traditionally, time has been viewed as a continuous parameter. However, in the discrete spacetime model, time is redefined as a series of discrete state transitions, with each moment corresponding to a specific entropy value. This characterization of time not only deepens our understanding of the nature of time but also offers a new perspective for interpreting the fundamental laws governing the evolution of the universe.

This study introduces a discrete Time-Entropy Mapping Model (1) where entropy is defined through multiplicative accumulation ($S = \prod_{i=1}^n m_i, i \in N$), rigorously satisfying entropy increase $\Delta S > 0$ while maintaining logarithmic consistency with conventional entropy. (2) The model establishes a fundamental spacetime-entropy mapping that unifies global time (universe transformations), local time (local space transformations), and thermodynamic evolution, providing a new computational entropy coordinate for physical simulations. (3) Key theoretical implications include deriving the arrow of time from Space Elementary Quantum (SEQ) dynamics, explaining light-speed invariance, and suggesting dark energy/matter as emergent SEQ ground-state phenomena. (4) A distinctive falsifiable prediction emerges: a measurable difference between positron and electron magnetic moments due to chiral SEQ coupling, testable in precision experiments. (5) The model provides three foundational constraints for computational physics simulations: thermodynamic conservation laws (energy/momentum), entropy-driven evolution ($\Delta S \geq 0$), and least-action path selection.

Key words: multiplicative entropy; space-time-entropy mapping; thermodynamic time arrow; entropy coordinate;

I. INTRODUCTION.

The thermodynamic arrow of time has yet to be rigorously defined, and conventional statistical entropy formulations fail to characterize the concrete process of entropy increase. This paper proposes a computable analytic expression for entropy within a discrete framework, explicitly describing:

- (1) The entropy growth process in closed systems,
- (2) The mapping relationship between entropy and time.

II. METHODS

1 Basic set:

- 1.1 The universe is expanding.
- 1.2 The universe operates under the Law of Energy Conservation and the Law of Entropy Increase.
- 1.3 The universe consists of Space Elementary Quanta (SEQ), forming a topologically homeomorphic 3D structure (adjacency relations remain preserved while individual SEQ energy states may vary). The distortion of light around black holes demonstrates that gravitational and electromagnetic field quanta are coupled, suggesting they originate from the same quantum field in different excited

states—leading to the SEQ hypothesis. The spacing and tension between adjacent SEQ can be modulated by gravitational or equivalent gravitational fields.

1.4 All field quanta and elementary particles represent different energy excitation states of SEQ, expressible as 3D dynamic structural matrices of SEQ.

1.5 SEQ possess a ground state energy (e.g., ground-state spin or vibrational modes). If ground-state spin chirality is fixed, this may explain parity violation. The ground-state energy of SEQ could also account for the cosmological constant in General Relativity.

This framework shows strong alignment with Loop Quantum Gravity theory.[1][2]

1.6 Adjacent SEQ maintain a dynamic equilibrium spacing. Perturbations from this equilibrium modify the SEQ's ground resonant frequency and induce associated compressive/tensile forces (origin of gravity and equivalent gravitation). The underlying response function is inherently nonlinear and asymmetric, potentially exhibiting complex behaviors including regional gradient variation. This elastodynamic formulation of spacetime exhibits remarkable conceptual synergy with Einstein's

original elastic continuum analogy [3], while providing a concrete realization of Zee's 'spring network' metaphor [4] through the discrete SEQ framework.

1.7 If matter truly traversed space, it would require modification of spacetime's adjacency relations. Yet black holes—despite their extreme mass—preserve local spacetime topology (as evidenced by smooth light bending). This implies that apparent particle motion must instead represent propagating excitations of spacetime itself, consistent with GR. The speed of light (c) constitutes the maximum excitation propagation rate in space.

1.8 Matter-Spacetime Unity: Matter is a part of space. General Relativity tells us that matter affects the metric of space, and the metric of space in turn determines the motion of matter along geodesics. In other words, matter and space influence each other. Algebraic module theory and category theory further reveal that two entities capable of mutual interaction must share part of the same algebraic structure. In categorical terms, relations are morphisms and interactions are structural. Expressed in physical language, this means that matter and space must share a common underlying basis—hence, matter and space-time are one.

2 The course of defining time:

2.1 SEQ serve as the electromagnetic wave conducting medium. Matter with mass and its motion are waves in this medium. In this framework, nothing truly moves through space - light speed c is the maximum conduction speed c , preventing velocity stacking beyond c . All physical phenomena correspond to specific energy state configurations, establishing SEQ as the universal substrate.

2.2 The universe's composition: Energy conservation and quantization imply a finite number N of SEQ, each with M possible energy states, allowing up to M^N transformations. These M energy states form an algebraic system incorporating translational, spinning and rotational operations. Energy conservation and entropy increase constraints reduce possible transformations significantly below M^N .

2.3 Time definition:

2.3.1 Let J be the possible universe transformations ($J \ll M^N$).

2.3.2 The Planck time (t_p) interval separates adjacent transformations as the minimal time unit.

2.3.3 Time's arrow follows entropy increase.

2.3.4 Transformations map non-bijectively to entropy values (k distinct values partition J transformations into K classes). Parallel

transformations share the same entropy values, but only one can occur. The entropy set maps to possible time values - each moment corresponds to one universe transformation. Non-uniform entropy increase means only a subset of possible time values actually occur.

2.3.5 Finite transformations ensure discrete, limited time.

Notably, such discrete evolutionary models find precursor in causal set theory's exploration of discrete spacetime dynamics [5], though our approach differs fundamentally in its entropic time construction."

3 Definition and analysis formula of entropy In this definition of entropy, energy state norms of the space elementary quantum are used to calculate specific closed space entropy value or the whole universe's entropy value. the entropy value can be defined by the cumulative multiplication of the norm of the energy state.

Entropy value of closed system $S = \prod_{i=1}^n m_i, i \in N$ ①

Energy of closed system $= \text{constant} = \sum_{i=1}^n m_i, i \in N$ ②

$S_{\max} \leq m_i^n$, When all m_i are equal or differ only by Planck's constant h

(Where m_i refers to the energy carried by the i th SEQ in a given closed system)

3.1 Energy transfer rules and triggering conditions:

Energy exchange occurs between adjacent SEQ (i, j) if and only if the following thermodynamic gradient exists: $m_i > m_j + h$, Energy transfer occurs only in discrete quanta of Planck's constant h , $m_i \rightarrow m_i - h$; $m_j \rightarrow m_j + h$ (Planck's constant: h)

3.2 Logarithmic Relation:

After logarithmic transformation, $\ln S$ aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.3 Proof of Spontaneous Entropy Increase

Theorem:

For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be $m_i = a, m_j = b$ ($a > b + h$);

The post-transfer entropy ratio is:

$S_{t2}/S_{t1} = (a-h)(b+h)/ab = 1 + h(a-b-h)/ab > 1$ (Planck's constant h)

3.4 Numerical Example: System States and Entropy Evolution

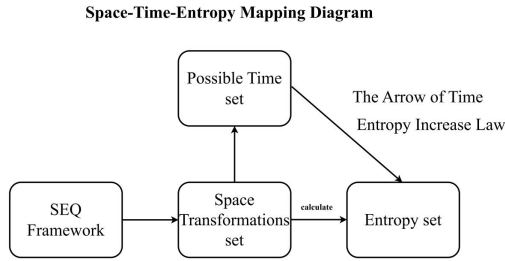
Table 1: Simplified Entropy Increase Demonstration

(table1: Simplified Entropy Increase Demonstration)

3.5 Entropy increase manifests through macroscopic interactions level, atomic/molecular level, QED processes, and microscopic particle restructuring (matrix disintegration/reorganization, radiation).

3.6 The entropy formulation= $\prod_1^n m_i; i \in N$ proposed herein operates not only at the fundamental SEQ level, but also admits coarse-graining for arbitrary massive many-body systems. This universality originates from the energy-mass equivalence which allows the cumulative product to naturally encode multi-body interactions. Crucially, this formulation provides first-principles computational constraints for discrete simulations across all scales.

At this point, a clear multiple mapping can be established:



Draw tool: draw.io [Computer software] <https://github.com/jgraph/drawio>

Diagram 1: Space-Time-Entropy Mapping Diagram
Drawing tool: draw.io [10]

3.7 The Spontaneous Entropy Increase is Causality. The existence of isentropic transformations implies that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, a certain dynamic freedom persists. The existence of isentropic transformations and different entropy-increasing pathways imply that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, with a certain dynamic freedom persists.

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System State	SEQ Energy Distribution $\sum_{i=1}^n m_i=12$	Entropy $\prod_{i=1}^n m_i$	Remarks
Initial non-equilibrium state	[3, 1, 5, 3]	45	-
Intermediate state	[3, 1, 4, 4]	48	-
Final state	[3, 2, 3, 4]	72	Due to adjacent energy transfer with minimal quanta h , this system cannot reach maximum entropy in this case

3.8 Why Analytic Entropy is Adopted: Within this model's framework, physical states at Planck-time scales exhibit deterministic characteristics. Although current experimental conditions cannot directly measure them, traditional statistical entropy can be reduced to analytic expressions at this scale. This perspective not only reveals the microscopic essence of statistical quantities but also provides a new analytic foundation for quantum thermodynamics—unifying macroscopic statistical behaviors with deterministic dynamics at the Planck scale.

Multiplicative analytic entropy corresponds to each step of energy homogenization in a system. With every step toward a more uniform energy distribution, the multiplicative analytic entropy increases, offering a clearer and more detailed characterization of the process compared to traditional statistical entropy.

3.9 Analysis of the Maximum Entropy Principle

In the SEQ model, the maximum entropy principle is manifested through the driving tendency of entropy increase: energy not only flows from a higher-energy SEQ to an adjacent lower-energy SEQ, but it also follows the path with the largest energy difference.

For example, consider an SEQ i , which is adjacent to two other SEQ j and k . The energies carried by nodes i , j , and k are A , B , and C respectively, where $A > B > C$. In this case, there are two possible energy transfer paths from SEQ i according to the principle of entropy increase: path $i \rightarrow j$ or path $i \rightarrow k$. We will analyze the entropy change for each path separately.

Under the constraint of energy conservation (i.e., $A + B + C = \text{constant}$), the entropy of the local system composed of these three SEQ before energy transfer is: $S = A \times B \times C$

Path $i \rightarrow j$: transferring energy to node j (which has relatively higher energy):

$$S' = (A - 1)(B + 1)C = (A - 1)(BC + C)$$

Path $i \rightarrow k$: transferring energy to node k (which has lower energy):

$$S'' = (A - 1)B(C + 1) = (A - 1)(BC + B)$$

Here, "1" represents one unit of Planck constant h . Since $B > C \Rightarrow BC + B > BC + C \Rightarrow S'' > S'$, the entropy increases more along the $i \rightarrow k$ path—that is, the path with the larger energy difference leads to a greater increase in entropy.

This deduction can be easily generalized to cases where the number of adjacent nodes is greater than two, so a detailed proof is omitted here.

However, it should be emphasized that the path with the maximum energy difference is not necessarily unique. Therefore, for states that have not yet occurred, the future still retains sufficient degrees of freedom — the evolution is not entirely deterministic.

3.9 In section 9.1.1, we will continue to discuss the decomposition of energy carried by SEQ, the relational expressions, and the supplementary description of the initial low-entropy state of the universe.

4 Analysis of Action

The dimension of the action quantity is consistent with that of Planck constant, so the action quantity can be the number of units of quantum energy (Planck constant).

The total number of unit quantum conduction energy of a physical process involves two parameters, one is the number n of SEQ involved in the physical process wave conduction, and the other is the conduction times k_i of the i th space elementary quantum involved in the conduction, in which the maximum number of conduction count k_i of all single SEQ is less than or equal to times of local spatial transformations in this process. h is Planck constant. Then the action amount can also be written as

$$\sum_{i=1}^n h k_i, i \in N \quad (3)$$

The law of the principle of least action reveals that the path selected by physics process is the path that the total amount of energy conduction involved SEQ is the least. It involves two parameters, as mentioned above. If the number of SEQ involved remains the same and the conduction times of each SEQ are the same, (1)Fermat's principle of the shortest time of optical path and (2)the principle of the steepest descent line can be directly derived, because in these two examples, the number of SEQ involved in the physical process and the rolling spherical rigid body in the steepest descent line remain the same, and the conduction times of each SEQ are also approximately the same, then time (the number of local space transformations) is the only variable for the calculation of action, so using time to divide the wave forms of different paths is equivalent to using

action to divide the wave forms of different paths. Time here refers to the transformation times of local space. From this point of view, we can probably understand why the analysis of action amount proposed in different periods is different, but it can explain some phenomena.

Comparison with Hamiltonian and Lagrangian:

Compared with the Lagrangian and Hamiltonian, the analytical expression herein does not explicitly contain a potential energy term. In subsequent sections (section 9 to section 10), it will be mentioned that any form of metric change in space leads to a decrease in SEQ resonance frequency - resonance frequency modulation.

Thus, the actual analytical expression in this chapter has already implicitly included the potential energy term. Conversely, under this model, the essence of the potential energy term in the analytical expressions of the Hamiltonian and Lagrangian is the modulation of the number of energy transmissions. Under this model, gravitational potential energy, electromagnetic potential energy, weak interaction potential energy, and strong interaction potential energy are essentially the elastic potential energy of tensor or twist distortion of space.

To elaborate further:

The absence of an explicit potential energy term in the analytical herein expression is compensated by the concept that any form of metric change in space results in a reduction of SEQ resonance frequency. This implies that the potential energy term is inherently embedded within the formulation via resonance frequency modulation.

The essence of the potential energy terms in both the Hamiltonian and Lagrangian formulations, under this model, can be understood as modulations in the frequency of energy transmission events.

Gravitational potential energy, electromagnetic potential energy, weak interaction potential energy, and strong interaction potential energy are all fundamentally manifestations of the elastic potential energy resulting from distortions in the spatial tensors or twists.

The essence of potential energy release is the reduction of spatial distortion, which is accompanied by an increase in SEQ resonance frequency.

A Brief Discussion on Causality:

Within the ontological framework of quantum spacetime, the law of causality is determined by three fundamental principles: the second law of entropy increase, the principle of maximum entropy path selection, and the principle of least action (minimum action path selection).

At the microscopic scale, the discrete evolution of SEQ network is governed by a Markov process — each state transformation at the Planck time unit depends only on the immediately preceding state. However, open possibilities are preserved through nonlinear energy redistribution.

At the macroscopic scale, the cumulative effect of such microscopic causal interactions gives rise to a quasi-deterministic picture under the multiple constraints of entropy increase, maximum entropy path selection, and the principle of least action.

Thus, the second law of entropy, the maximum entropy path principle, and the minimum action path principle together constitute the foundational logic of cosmic causality: the former provides the arrow of irreversibility in evolution, while the latter two ensure the optimality of path selection. Together, they determine a unified causal network spanning from the quantum microscopic level to the observable macroscopic world.

However, since the path of least action is not necessarily unique, nor is the maximum entropy path, and the rate of entropy increase also has degrees of freedom (as will be discussed in Chapter 13 in relation to the duality between spatial deformation and SEQ resonance frequency), the future still retains uncertainty — it is not strictly deterministic.

5 Local time , the proper time and relative time in Relativity

5.1 local time. As previously established, global time is defined by the transformations of the universe. This work introduces local time as an operational concept that: (1) provides correspondence with special relativistic time notions; and (2) enables precise specification of time scales for localized physical processes. Crucially, any measurable time parameter in physical calculations necessarily corresponds to transformations within a specific local space. This operational concept is designated as **local time**.

The local space scope must be unambiguously specified: either as the SEQ network along the physical process path or the connected region within the observer's light cone. This distinction mirrors the complementarity between Feynman's path-integral formulation and relativistic theory, avoiding conceptual confusion in prior works.

Local time: it can specify the time set corresponding to the transformations within a specific spatial range. In fact, from the perspective of this framework, the

existing equations with t parameter in textbooks are actually the local time by default.

One example: the clock slowness effect in Relativity manifests as different transformation times in different local spaces.

Every local space transformation constitutes a part of the universal evolution. Global time progression (ΔT_{global}) does not necessitate synchronous local transformations, the state matrix of a given local space may remain invariant despite cosmic-scale changes ($\Delta \tau_{\text{local}} \approx 0$) when this local space undergoes no state transition.

5.2 The proper time in Relativity is related to local transformations count of physics process entities. (Section 6.5 present the physical mechanism underlying proper time dilation in GR.

5.3 Understanding on relative time in Special Relativity theory: The time perception of physical processes across distinct reference frames fundamentally corresponds to the observation of transformation counts. The observation time discrepancy between frames derives from the accumulated difference in their frame SEQ transformation counts. An observer measures another frame's time evolution through the differential transformation count ΔN , while he can't perceive their own transformation count N_0 .

- Key Distinction from GR Effects

While the mathematical derivation process aligns with standard special relativity textbooks—replacing continuous spacetime metrics with discrete counting operations—the physical interpretation differs substantially in conceptualization:

- SR Effects as Perceptual Phenomena

Time dilation and length contraction emerge purely as observer-dependent measurement consequences

Originate from information transmission constraints via discrete light-signal counting

- Contrast with GR Mechanisms

Gravitational time dilation involves actual deformation of the SEQ network's transformation frequency

Equivalence principle effects require space compression as well (see [12])

5.4 Physical Meaning of Planck Time

In this framework, the Planck time (Δt_P) corresponds to the fundamental harmonic period of the SEQ network in its equilibrium state—the duration for a

complete energy transfer (or harmonic oscillation period) between adjacent SEQ. This period defines the theoretical minimum transformation cycle (i.e., maximum transformation frequency) of the universe as a whole.

While gravitational fields or equivalent interactions can locally modulate the harmonic frequency by deforming the SEQ network (e.g., compression/stretching, as in gravitational time dilation), the global maximum theoretical transformation frequency remains anchored by in the equilibrium state. However, due to the omnipresence of gravitational effects, the empirically observable maximum transformation frequency of space may marginally less than this theoretical limit.

6 Phenomenological consistency checks

6.1 Why can't the speed of light stack up?

As established above, the speed of light (c) constitutes the maximum excitation propagation rate in space, wherein all observed motion fundamentally represents state transitions within this immutable lattice rather than physical traversal through space.

6.2 Uncertainty Relation and Wave-particle Duality.

The Uncertainty Principle naturally arises from wave propagation through discrete SEQ: precise position inherently limits determination of conduction speed (wave dynamics), and vice versa.

Wave-particle Duality. The wave nature is fundamental, while the particle nature emerges from the discreteness of spacetime itself.

6.3 Double-slit experiment. [6][7]

Explanation: Based on this framework, electron's conducting in the space conforms some probability function, and really causes multi-path oscillating in space, and these oscillating can be accumulated. Electron excitation causes space oscillating both of the two slits at the sides of excitation source, The proposed framework predicts the emergence of interference fringes, as the electron excitation induces coherent oscillations across multiple SEQ conduction paths. Even in case of one-slit experiment, when the accumulated Wavelets cross the slit, slit's Unsmooth edges may produce different paths and generate interference that could be observed if the sensor is sensitive enough and the slit edge's form can meet the Coherent Condition.

Note: While this framework strictly enforces time ordering along entropy increase, it currently does not provide an interpretation for delayed-choice experiments. This remains an open question requiring further development in this framework.

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6.4 Non-conservation of parity

With the assumption above, if SEQ ground state spin chirality is fixed, that could be one possible explanation of the non conservation of parity.

Is dark matter potentially explained by high-density SEQ ground-state clusters under gravity? And is the ground state energy carried by the SEQ the so-called dark energy?

6.5 Conjecture on Muon Decay experiment[8]

An object's motion (velocity/acceleration) deforms the SEQ network via:(1)stress-energy dependent spacing variation,(2) strain-modulated transformation frequencies , and (3) GR-consistent effects (velocity-dependence mirrors SR, acceleration induces curvature). Particle decay results from destabilization of their 3Dstructure matrices when interaction forces (EM/weak/strong) fail to maintain equilibrium.

7 A prediction of a difference in the magnetic moments of the positron and electron(falsifiable)

Given the isotropic nature of the electric field generated by electrons, this framework hypothesizes that electrons possess an Spatially Symmetric structure composed of SEQ. Under the SEQ framework, all charged microscopic particles including electrons and quarks possess dynamic three-dimensional structures that preserve spherical symmetry in space.

If a slight difference is measured between the magnetic moment of the positron and that of the electron, it would strengthen the credibility of the hypothesis that the SEQ has a ground-state spin. This difference arises because the positron's structural matrix rotates with opposite chirality to the electron's, resulting in distinct coupling configurations with the SEQ's fixed-chirality ground-state spin. Based on this, it can be inferred that the magnetic moments of the positron and electron should exhibit a slight discrepancy. A precision measurement of the positron magnetic moment, currently underway by the Fan-Myers-Sukra-Gabrielse collaboration , could test two key hypothesis of the SEQ framework: the fixed chirality of SEQ ground-state spin and electron's substructure.[9]

III. RESULTS

1.The study successfully establishes a time-entropy mapping model within the discrete spacetime framework, where each temporal state corresponds to a unique entropy value. This mapping ensures that entropy either increases or remains constant over time, providing a clear thermodynamic arrow of time.

2. However, the mapping is not bijective.
3. The model has undergone consistency checks against most key phenomena, confirming its physical plausibility, while proposing a falsifiable criterion.

IV. DISCUSSION

While the present model does not yet incorporate electromagnetic interactions or deeper algebraic structures, it establishes a foundational framework for such future developments. We envision that further development using an algebra system and discrete differential geometry could refine this framework, ultimately achieving better alignment with both general relativity and the Standard Model.

Within this framework, gravitational interaction emerges as a translational operation that modulates the inter-SEQ spacing and consequently alters the stress-energy distribution; the emergent gravitational field of mass exhibits spherical symmetry due to SU(3) color charge distribution[11]. Serving as the

symmetry-preserving mechanism at quantum scale, While this manuscript focuses on space-time-entropy mapping,

V. CONCLUSIONS

This work presents a theoretical framework whose mathematical formulation, while deliberately minimalist, yields tractable derivations and falsifiable predictions. The model demonstrates particular utility in elucidating the fundamental relationship between temporal evolution and entropy, with immediate applications to computational simulations through the embedding constraints of thermodynamic laws and least-action path selection offering an entropy coordinate for computer evolution simulations.

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